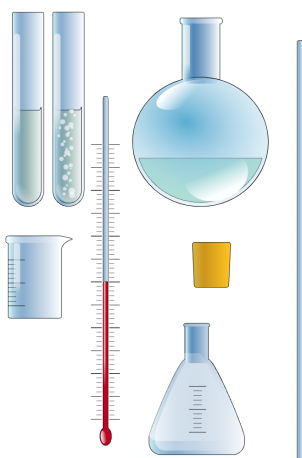




UNIVERSITY INTERSCHOLASTIC LEAGUE

Science

Invitational B • 2024



GENERAL DIRECTIONS:

- **DO NOT OPEN EXAM UNTIL TOLD TO DO SO.**
- Contestants may take up to two hours to complete the contest. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet. Write clearly and legibly!
- You may place as many notations as you desire anywhere on the test paper but not on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- The back two pages of this test include a copy of the periodic table of the elements, as well as listings of other scientific relationships. You may use this information during the contest and may detach the back page from the test if you wish.
- A simple scientific calculator is sufficient for the high school Science contest. **The UIL provides a list of approved calculators that meet the criteria for use in the Science contest. No other calculators are permitted during the contest.** The Science Contest Approved Calculator List is available in the current Science Contest Handbook and on the UIL website. Contest directors will perform a brief visual inspection to confirm that all contestants are using only approved calculators. Each contestant may use up to two approved calculators during the contest.

- B01. Which of the following statements is not true about biological membranes?
- A) The hydrogenation of fatty acids influences the fluidity.
 - B) The hydrophilic heads of phospholipids are always on the outside of the membrane with the fatty acid tails pointing inward.
 - C) Cholesterol embedded within the membranes helps control fluidity.
 - D) Having only single covalent bonds between the carbons of the fatty acid tails makes the membrane more fluid.
 - E) Plasma membrane, nuclear membrane, and mitochondrial membranes are all examples of biological membranes.
- B02. A single base change that results in a conversion of the amino acid from phenylalanine to histidine is called a
- A) nonsense mutation.
 - B) silent mutation.
 - C) coding mutation.
 - D) frameshift mutation.
 - E) missense mutation.
- B03. Signaling through the use of hormones occurs via the _____ system.
- A) integumentary
 - B) nervous
 - C) endocrine
 - D) digestive
 - E) respiratory
- B04. During chemiosmosis of aerobic respiration, which molecule is moving across the membrane, thus releasing energy to generate ATP?
- A) oxygen
 - B) ATP
 - C) potassium
 - D) hydrogen ions
 - E) carbon dioxide
- B05. Which of the following is mismatched with its transport mechanism?
- A) Channel proteins moving substances from high to low concentration = facilitated diffusion
 - B) Movement of lipid-soluble molecules from high to low concentration directly through the membrane = active transport.
 - C) ATP hydrolysis while moving a sugar from low to high concentration through a transport protein = active transport.
 - D) Movement of substances across the membrane from high to low without a transporter = simple diffusion
 - E) Osmosis of water = passive transport
- B06. Assuming Hardy-Weinberg equilibrium, a population has 43% who are homozygous dominant. What is the frequency of the recessive allele?
- A) 0.119
 - B) 0.344
 - C) 0.451
 - D) 0.570
 - E) 0.656
- B07. The enzyme that replicates a majority of the nucleic acid during DNA replication is
- A) DNA polymerase I
 - B) Helicase
 - C) DNA polymerase III
 - D) Primase
 - E) RNA polymerase III
- B08. Which of the following would be the best technique, relative to the other choices, to precisely edit the base of a known DNA sequence?
- A) CRISPR/Cas9
 - B) Western blot
 - C) Northern blot
 - D) Southern blot
 - E) Transduction

- B09. The mechanism of evolution that states organisms that are more adapted to their environment are more likely to be successful and pass on the genes to their offspring is called
- natural selection.
 - speciation.
 - gene flow.
 - adaptation.
 - Lamarckism.
- B10. An excess of calcium in the blood is called
- hyperproteinemia.
 - hyperkalemia.
 - hypercalcemia.
 - hypernatremia.
 - metabolic alkalosis.
- B11. Which of the following is not an example of microevolution?
- Bacteria developing antibiotic resistance.
 - Mosquitos developing resistance to a pesticide, such as DDT.
 - Viruses becoming resistance to antiviral medications.
 - The emergence of two different species after geographic isolation.
 - None of the above are examples of microevolution.
- B12. Yeast are single-celled organisms that belong to Domain
- Prokarya.
 - Archaea.
 - Bacteria.
 - Fungi.
 - Eukarya.
- B13. Mammals all belong to the same
- class.
 - order.
 - family.
 - genus.
 - species.
- B14. In November 2023, the Centers for Disease Control and Prevention issued alerts for several food items due to possible contamination with *Salmonella*. Which of the following was not part of the alerts?
- fresh diced onions
 - dry dog food
 - cantaloupe
 - peaches
 - oysters
- B15. All of the following are *Enterobacteriaceae* except
- Klebsiella pneumoniae*.
 - Pseudomonas aeruginosa*.
 - Shigella dysenteriae*.
 - Escherichia coli*.
 - Salmonella enterica*.
- B16. Which of the following blood types is not a possibility from the following genetic cross?
- $$I^A i \times I^A I^B$$
- A
 - B
 - AB
 - O
 - All of the above are possibilities.
- B17. Examine the chromosome structures below. Which event occurred in the mutation? Note: the “*” represents the centromere.
- Normal: ABCD*EFGHI
Mutant: ADCB*EFGHI
- inversion
 - translocation
 - deletion
 - duplication
 - reciprocal translocation

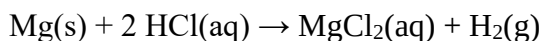
- B18. In the sulfur cycle, sulfates (SO_4^{2-}) are reduced to hydrogen sulfide through
- A) aerobic respiration.
 - B) microbial anaerobic respiration.
 - C) assimilation.
 - D) fermentation.
 - E) microbial oxidation reactions.
- B19. Rubisco
- A) is an enzyme involved in DNA replication.
 - B) converts a DNA sequence into RNA.
 - C) releases carbon dioxide during the Krebs cycle.
 - D) is a component of reaction centers in the light-harvesting reactions of photosynthesis.
 - E) catalyzes carbon dioxide fixation in photosynthesis.
- B20. Proteins that hold sister chromatids together until the right moment of separation during mitosis are called
- A) shugoshin.
 - B) histones.
 - C) cohesins.
 - D) kinetochores.
 - E) adhesins.

- C01. If 4.00×10^{23} atoms of helium are added to a latex balloon, what is the mass of the helium inside the balloon?



- A) 0.266 grams
- B) 0.900 grams
- C) 1.13 grams
- D) 2.66 grams
- E) 4.22 grams

- C02. For the reaction



which of the following statements is true at STP?

- A) 2 moles of HCl produce 1 L of H_2 gas
- B) 1 gram of Mg produces 1 gram of H_2 gas
- C) 2 L of HCl produces 1 mole of H_2 gas
- D) 24.31 grams of Mg produces 1 gram of H_2 gas
- E) 24.31 grams of Mg produces 22.4 L of H_2 gas

- C03. When barium forms the ionic compound BaCl_2 , how many electrons are in the barium ion?

- A) 34 B) 54 C) 56 D) 58 E) 137

- C04. Which pair of elements below is most likely to form a covalent bond?

- A) magnesium and fluorine
- B) carbon and hydrogen
- C) copper and iron
- D) aluminum and chlorine
- E) None of these combinations would form a covalent bond

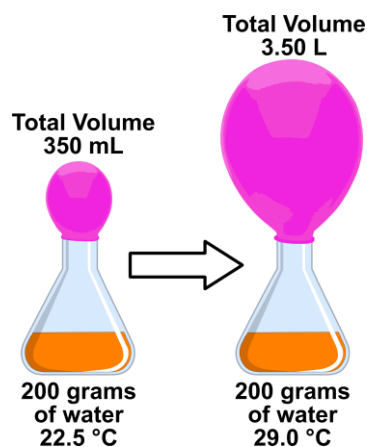
- C05. A sample of gas in a rigid container has a pressure of 2.0 atm and is heated from 50°C to 100°C . Which of these describes the new pressure?

- A) The new pressure would be 4.0 atm.
- B) The new pressure would be 1.0 atm.
- C) The new pressure would be more than 2.0 atm but less than 4.0 atm.
- D) The new pressure would be less than 2.0 atm but more than 1.0 atm.
- E) The pressure would be unchanged.

- C06. Which of these liquids would have the highest boiling point?

- A) CH_4
- B) CH_3F
- C) CH_3Cl
- D) CH_3F_2
- E) CH_3OH

- C07. A chemical reaction was carried out in an aqueous solution at 1 atm pressure in a balloon-covered flask and the following changes were observed.



How much heat was given off (–) or absorbed (+) by this reaction?

- A) –5439 J
- B) +5439 J
- C) –1300 J
- D) +1300 J
- E) +4184 J

- C08. What would the boiling point of the solution be if you dissolved 20.0 grams of NaCl in 250 grams of water?
- A) 100.7 °C
B) 101.4 °C
C) 102.2 °C
D) 104.5 °C
E) 105.1 °C
- C09. Zn^{2+} reacts with aqueous ammonia to form the tetraamminezinc(II) ion, $\text{Zn}(\text{NH}_3)_4^{2+}$. What is the equilibrium expression for the formation of this complex ion?
- A) $K_f = \frac{[\text{Zn}^{2+}][\text{NH}_3]^4}{[\text{Zn}(\text{NH}_3)_4^{2+}]}$
B) $K_f = \frac{[\text{Zn}(\text{NH}_3)_4^{2+}]}{[\text{Zn}^{2+}][\text{NH}_3]^4}$
C) $K_f = \frac{[\text{Zn}(\text{NH}_3)_4^{2+}]}{[\text{Zn}^{2+}]}$
D) $K_f = \frac{[\text{Zn}(\text{NH}_3)_4^{2+}]}{[\text{Zn}^{2+}]^4[\text{NH}_3]}$
E) $K_f = \frac{[\text{Zn}^{2+}]^4[\text{NH}_3]}{[\text{Zn}(\text{NH}_3)_4^{2+}]}$
- C10. What is the pH of an 8.95×10^{-6} M solution of NaOH?
- A) 5.05
B) 6.33
C) 7.09
D) 8.21
E) 8.95
- C11. The K_{sp} for $\text{Ni}(\text{OH})_2$ is 2.8×10^{-16} . What is the concentration of OH^- ions in a saturated solution of $\text{Ni}(\text{OH})_2$?
- A) 4.1×10^{-6} M
B) 2.4×10^{-8} M
C) 3.2×10^{-5} M
D) 1.3×10^{-5} M
E) 8.2×10^{-6} M
- C12. In which of these chemical reactions is hydrogen being reduced?
- A) $\text{H}_2(g) + 2\text{O}_2(g) \rightarrow 2\text{H}_2\text{O}(g)$
B) $2\text{HCl}(aq) + \text{Zn}(s) \rightarrow \text{H}_2(g) + \text{ZnCl}_2(aq)$
C) $\text{HCl}(aq) + \text{NaOH}(aq) \rightarrow \text{NaCl}(aq) + \text{H}_2\text{O}(\ell)$
D) $\text{CH}_4(g) + 2\text{O}_2(g) \rightarrow \text{CO}_2(g) + 2\text{H}_2\text{O}(\ell)$
E) $2\text{H}_2(g) + \text{C}(s, \text{graphite}) \rightarrow \text{CH}_4(g)$
- C13. The rate law for an aqueous reaction that is first order in A and B and second order overall is $\text{rate} = k[\text{A}][\text{B}]$. What are the units on the rate law constant k ?
- A) M/s
B) M·s
C) s/M
D) $\text{M}^{-1} \cdot \text{s}^{-1}$
E) s/M^2
- C14. A 100 gram sample of KBr contains
- A) equal masses of potassium and bromine
B) 39.1 grams of potassium and 79.9 grams of bromine
C) more than twice as many grams of bromine as potassium
D) more than twice as many grams of potassium as bromine
E) More than one of the above answer choices is correct
- C15. Which of these forms of electromagnetic radiation has the longest wavelength?
- A) Microwaves
B) Infrared
C) Visible light
D) Gamma radiation
E) Ultraviolet light

C16. What is the mass density of a sample of fluorine gas at STP?

- A) 1.70 g/L
- B) 0.85 g/L
- C) 0.045 g/L
- D) 1.00 g/L
- E) 1.35 g/L

C17. The homework assignment said to write a valid set of quantum numbers for the outermost electron in a nitrogen atom. Your friend turned in his answer and got zero credit, but he doesn't know why. What is wrong with his answer, shown here:

**Possible quantum numbers
for the outermost electron
in a nitrogen atom:**

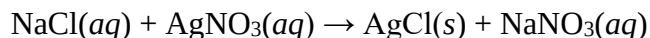
$$n = 2, \ell = 2, m_{\ell} = 0, m_s = -\frac{1}{2}$$

- A) Nitrogen has three electrons in the outermost subshell, so $n = 3$.
- B) m_{ℓ} can be positive or negative but cannot be 0
- C) m_s has to have the same sign as m_{ℓ} , and since his $m_{\ell} = 0$, m_s must also be 0.
- D) Since $\ell = 2$, n must be at least 3 because $\ell = 0, 1, 2, \dots, n - 1$
- E) $n = 2$ and $\ell = 2$ would be the $2d$ subshell and there is no $2d$ subshell

C18. Which of these properties of a gas molecule contributes most to the gas behaving non-ideally?

- A) The number of different elements that make up the gas molecule
- B) The number of hydrogen atoms that are found in the gas molecule
- C) The molar mass of the gas molecule
- D) The polarity of the gas molecule
- E) The geometrical shape of the gas molecule

C19. A student dissolves 7.50 grams of NaCl in a beaker containing 100 mL of water and dissolves 30.0 grams of AgNO₃ in a different beaker containing 300 mL of water, and then combines the two solutions. The following precipitation reaction occurs:



The student then isolates and dries the solid product. If the yield is 100%, what is the mass of the solid product?

- A) 7.50
- B) 15.0
- C) 18.4
- D) 12.7
- E) 10.9

C20. How is the actual yield in a chemical reaction calculated?

- A) From the moles of the limiting reactant in the reaction.
- B) From the moles of the excess reactant in the reaction.
- C) From the moles of the excess reactant remaining after the reaction is over.
- D) From the total number of moles of reactants used in the reaction.
- E) Actual yield is measured experimentally and cannot be calculated from the moles of reactants used.

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

T_{mp}	$= 0^{\circ}\text{C}$
T_{bp}	$= 100^{\circ}\text{C}$
c_{ice}	$= 2.09 \text{ J/g}\cdot\text{K}$
c_{water}	$= 4.184 \text{ J/g}\cdot\text{K}$
c_{steam}	$= 2.03 \text{ J/g}\cdot\text{K}$
ΔH_{fus}	$= 334 \text{ J/g}$
ΔH_{vap}	$= 2260 \text{ J/g}$
K_{f}	$= 1.86^{\circ}\text{C}/m$
K_{b}	$= 0.512^{\circ}\text{C}/m$

$$\begin{aligned} R &= 0.08206 \text{ L}\cdot\text{atm/mol}\cdot\text{K} \\ R &= 8.314 \text{ J/mol}\cdot\text{K} \\ R &= 62.36 \text{ L}\cdot\text{torr/mol}\cdot\text{K} \\ e &= 1.602 \times 10^{-19} \text{ C} \\ N_{\text{A}} &= 6.022 \times 10^{23} \text{ mol}^{-1} \\ k &= 1.38 \times 10^{-23} \text{ J/K} \\ h &= 6.626 \times 10^{-34} \text{ J}\cdot\text{s} \\ c &= 3.00 \times 10^8 \text{ m/s} \\ R_{\text{H}} &= 2.178 \times 10^{-18} \text{ J} \\ m_{\text{e}} &= 9.11 \times 10^{-31} \text{ kg} \\ \mathcal{F} &= 96,485 \text{ C/mol e}^{-} \\ 1 \text{ amp} &= 1 \text{ C/sec} \\ 1 \text{ mol e}^{-} &= 96,485 \text{ C} \end{aligned}$$
$$1 \text{ L}\cdot\text{atm} = 101.325 \text{ J}$$

P01. According to Guillen, Jakob and Johann Bernoulli struggled with a paper in which Leibniz described the mathematics of calculus. Jakob eventually understood the paper, which hinged on a new idea called...

- A) the derivative
- B) the Riemann sum
- C) the limit
- D) the differential
- E) the infinitesimal

P02. According to Guillen, Johann Bernoulli began working on an idea regarding a substance possessed by objects that were to some degree animated. He called this substance *vis viva*. Today we know it as ...

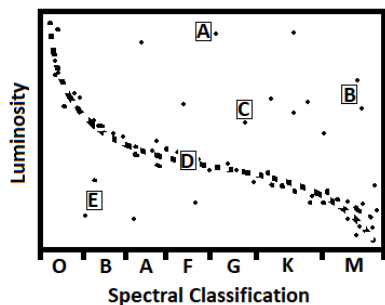
- A) force
- B) velocity
- C) momentum
- D) energy
- E) heat

P03. According to Guillen, Nikolai Zhukovsky utilized Daniel Bernoulli's fluid-flow equation to explain how ...

- A) windmills pump water.
- B) airplanes fly.
- C) ocean current flow.
- D) storm cells form.
- E) buoyancy works.

P04. Consider a star similar in mass to our Sun. Towards the end of its life, as the star evolves beyond its red giant phase, it will move across the Hertzsprung-Russell diagram along what is called the 'horizontal branch.' It is along the horizontal branch that many variable stars, such as RR Lyrae stars, exist. Approximately where on the Hertzsprung-Russell diagram would a horizontal branch star be located?

- A) point A
- B) point B
- C) point C
- D) point D
- E) point E



P05. Which result for the following calculation has the correct number of significant figures?

$$x = \frac{13.3 - 9.55}{56.1}$$

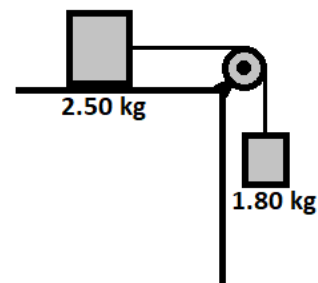
- A) 0.1
- B) 0.07
- C) 0.067
- D) 0.0668
- E) 0.06684

P06. A dog toy is thrown straight up with an initial velocity of 9.65m/s. The toy hits the ceiling, which is located 3.35m above where the toy was thrown. How fast is the toy moving when it hits the ceiling?

- A) 5.24 m/s
- B) 6.30 m/s
- C) 7.76 m/s
- D) 8.53 m/s
- E) 9.29 m/s

P07. A 1.80kg box is attached to a rope that passes over a pulley and connects to a 2.50kg crate (as shown). The crate is sitting on a frictionless horizontal tabletop, and the pulley is also massless and frictionless. Once released, the box and crate are free to move. What, then, is the acceleration of the two objects?

- A) 2.74 m/s²
- B) 3.81 m/s²
- C) 4.10 m/s²
- D) 5.70 m/s²
- E) 7.06 m/s²



P08. A nail with a mass of 32.0g passes clean through a block of wood. The nail is initially moving at 80.0m/s. After passing through the block of wood, the nail has slowed to 43.0m/s. The block of wood is initially at rest and has a mass of 215g. What is the speed of the block of wood immediately after the nail passes through it?

- A) 18.3 m/s
- B) 11.9 m/s
- C) 6.40 m/s
- D) 5.51 m/s
- E) 1.16 m/s

P09. A horizontal merry-go-round has a moment of inertia of 8.60kgm^2 and an initial angular velocity of 36.0rad/s . A tree root that is pressed up against the merry-go-round, exerts a torque on the merry-go-round that causes it to slow to a stop after 4.00 complete revolutions. What is the magnitude of the torque exerted by the tree root?

- A) 151 Nm
- B) 222 Nm
- C) 310 Nm
- D) 443 Nm
- E) 887 Nm

P10. A steel cable is used to hold up a suspension bridge. The cable has a diameter of 18.0cm and is 450.0m in length. During a windstorm, the cable is subjected to an additional pulling force of 8600.0N . By how much does the cable stretch due to the force of the windstorm?

Young's modulus for steel is $2.0 \times 10^{11}\text{ Pa}$.

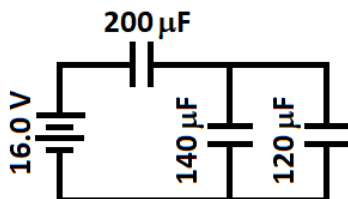
- A) 2.3 mm
- B) 0.76 mm
- C) 0.60 mm
- D) 0.38 mm
- E) 0.19 mm

P11. A diatomic ideal gas starts at a pressure of 125kPa and a volume of 0.355 liters . The gas expands adiabatically to a volume of 0.610 liters . What is the pressure of the gas after the expansion? Note: The heat capacity ratio for a diatomic ideal gas is 1.40.

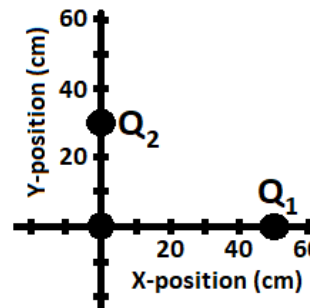
- A) 102 kPa
- B) 84.9 kPa
- C) 72.7 kPa
- D) 58.6 kPa
- E) 44.4 kPa

P12. For the capacitor circuit shown, determine the charge stored on the $140\mu\text{F}$ capacitor?

- A) $2790\mu\text{C}$
- B) $2240\mu\text{C}$
- C) $1810\mu\text{C}$
- D) $1270\mu\text{C}$
- E) $974\mu\text{C}$



P13. Two charges are placed on a coordinate grid as shown. The first charge, $Q_1 = +25.0\mu\text{C}$, is located at $(50.0\text{cm}, 0.0)$ and the second charge, $Q_2 = -12.0\mu\text{C}$, is located at $(0.0, 30.0\text{cm})$. What is the total electric potential, V , at the origin $(0.0, 0.0)$ due to the two charges?

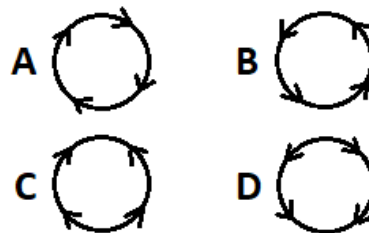


- A) 89.9 kV
- B) 576 kV
- C) 809 kV
- D) 1.50 MV
- E) 2.10 MV

P14. A charged particle with a velocity of $2.40 \times 10^5\text{m/s}$ enters an area in which there is a magnetic field. The magnetic field has a strength of 3.60mT and is oriented perpendicular to the velocity of the particle. The particle has the same charge as a proton, and in the field region it traces out a circular path with a diameter of 246.0cm . What is the mass of the particle as compared to the mass of a proton?

- A) $1.77 m_p$
- B) $2.96 m_p$
- C) $4.43 m_p$
- D) $5.91 m_p$
- E) $7.09 m_p$

P15. A circle of wire is placed horizontally on a table. A bar magnet is held vertically above the circle, with the North pole of the magnet pointed downward. The magnet is quickly pulled directly upwards, away from the circle of wire. In which direction, as seen from above, does the induced current flow in the circle of wire?



- E) There is no current induced in the wire.

P16. A quarter lies at the bottom of a pond, 45.0cm below the water's surface. How far under the surface does the quarter *appear* to be located (in other words: where is the image of the quarter located)? The water's surface is flat and smooth; the index of refraction of water is 1.33.

- A) 19.1cm below the surface
- B) 25.4cm below the surface
- C) 33.8cm below the surface
- D) 45.0cm below the surface
- E) 59.9cm below the surface

P17. An electron has a velocity of 6.25×10^6 m/s. What is the de Broglie wavelength of this electron?

- A) 0.0185 nm
- B) 0.0370 nm
- C) 0.0720 nm
- D) 0.106 nm
- E) 0.116 nm

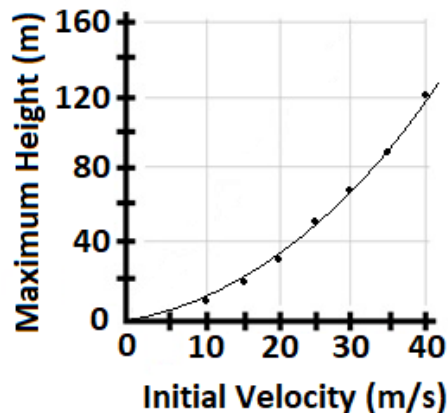
P18. An atom of ${}_{102}^{266}\text{No}$ undergoes the following radioactive decays:

$\alpha, \beta^-, \alpha, \alpha, \beta^-, \alpha, \alpha, \gamma$.

What is the daughter isotope resulting from this series of radioactive emissions?

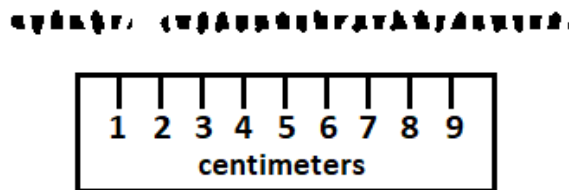
- A) ${}_{93}^{245}\text{Np}$
- B) ${}_{93}^{246}\text{Np}$
- C) ${}_{94}^{245}\text{Pu}$
- D) ${}_{94}^{246}\text{Pu}$
- E) ${}_{95}^{246}\text{Am}$

P19. You are on an alien planet doing a physics experiment. In the experiment, a ball is launched directly upward, and the maximum height reached by the ball is measured as a function of the launch velocity. The data are plotted below. What is the acceleration due to gravity on the alien planet?



- A) 3.2 m/s^2
- B) 6.6 m/s^2
- C) 9.8 m/s^2
- D) 13 m/s^2
- E) 17 m/s^2

P20. You direct a Helium-Neon laser with a wavelength of 633nm onto a double slit apparatus, producing an interference pattern on a screen that is located 2.40m from the double slit. The pattern that you observe is shown below, along with a ruler for scale. Based on the results, what is the separation distance of the two slits in the double slit apparatus?



- A) 3.19 mm
- B) 1.28 mm
- C) 0.64 mm
- D) 0.32 mm
- E) 0.16 mm

Physics

Useful Constants

quantity	symbol	value
Free-fall acceleration	g	9.80 m/s^2
Permittivity of Free Space	ϵ_0	$8.854 \times 10^{-12} \text{ C}^2/\text{Nm}^2$
Permeability of Free Space	μ_0	$4\pi \times 10^{-7} \text{ Tm/A}$
Coulomb constant	k	$8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$
Speed of light in a vacuum	c	$3.00 \times 10^8 \text{ m/s}$
Fundamental charge	e	$1.602 \times 10^{-19} \text{ C}$
Planck's constant	h	$6.626 \times 10^{-34} \text{ Js}$
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.67265 \times 10^{-27} \text{ kg}$ 1.007276 amu
Neutron mass	m_n	$1.67495 \times 10^{-27} \text{ kg}$ 1.008665 amu
Atomic Mass Unit	amu	$1.66 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
Gravitational constant	G	$6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$
Stefan-Boltzmann constant	σ	$5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$
Universal gas constant	R	$8.314 \text{ J/mol} \cdot \text{K}$ $0.082057 \text{ L} \cdot \text{atm/mol} \cdot \text{K}$
Boltzmann's constant	k_B	$1.38 \times 10^{-23} \text{ J/K}$
Speed of Sound (at 20°C)	v	343 m/s
Avogadro's number	N_A	$6.022 \times 10^{23} \text{ atoms/mol}$
Electron Volts	eV	$1.602 \times 10^{-19} \text{ J/eV}$
Distance Conversion	miles $\rightarrow$ meters	1.00 mile = 1609 meters
Rydberg Constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
Standard Atmospheric Pressure	1 atm	$1.013 \times 10^5 \text{ Pa}$
Density of Pure Water	ρ_{water}	1000.0 kg/m^3